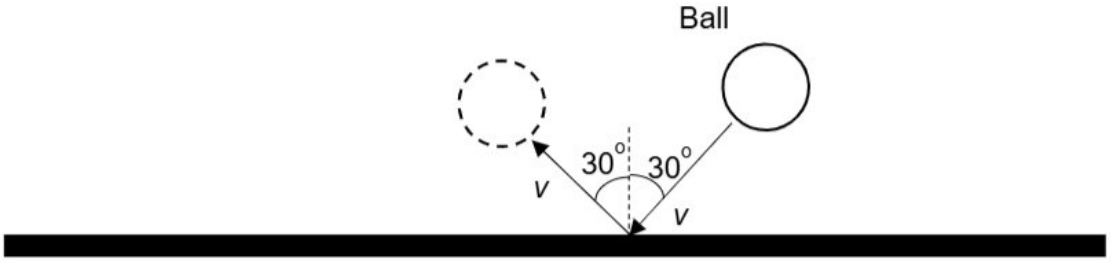
11 A snooker ball of mass m travels at a speed v towards the side of a snooker table as shown below. The ball strikes the side of the table at an angle of 30° to the normal and moves off with the same speed at an angle of 30°. The time of contact between the ball and the side is Δt. What is the expression for the average force acting on the ball?



TOP VIEW

A $\dfrac{mv}{\Delta t}$

B $\dfrac{mv\cos 30^\circ}{\Delta t}$

C $\dfrac{2mv\cos 30^\circ}{\Delta t}$

D $\dfrac{2mv\sin 30^\circ}{\Delta t}$